

MediBuddy Insurance Data Analysis Report

This report presents an exploratory data analysis (EDA) of MediBuddy insurance data to understand the factors influencing medical insurance charges.

1. Introduction

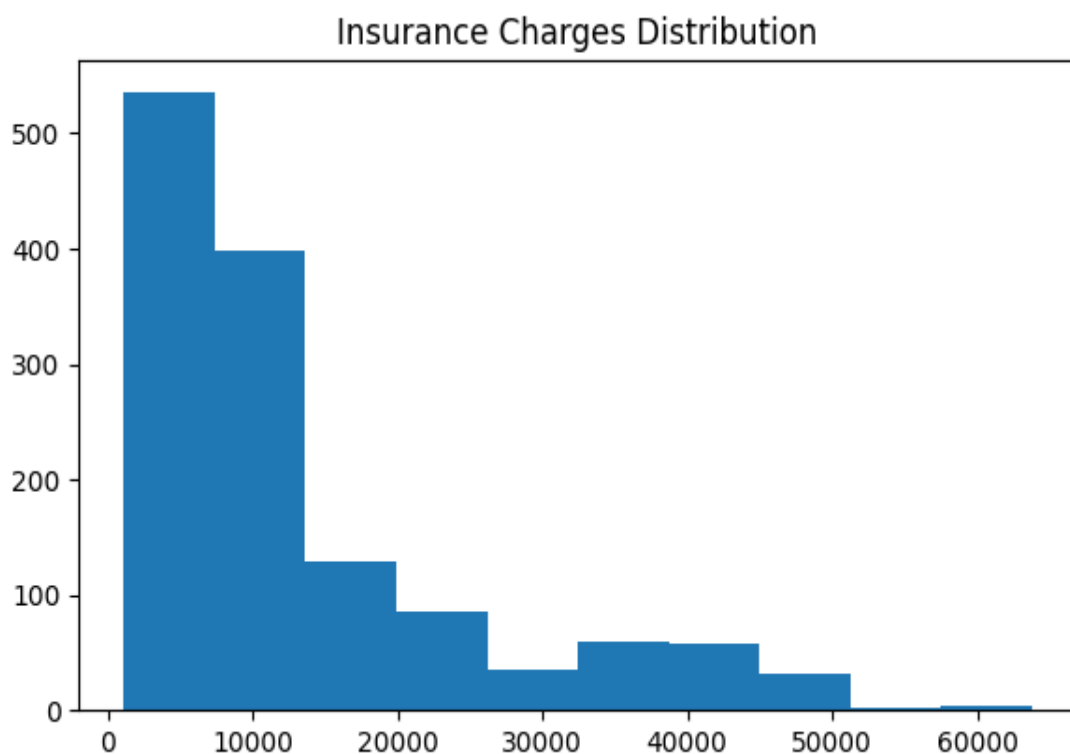
MediBuddy provides health insurance services to a wide customer base. Understanding factors such as age, BMI, smoking habits, and dependents is essential for effective pricing and risk management.

2. Dataset Description

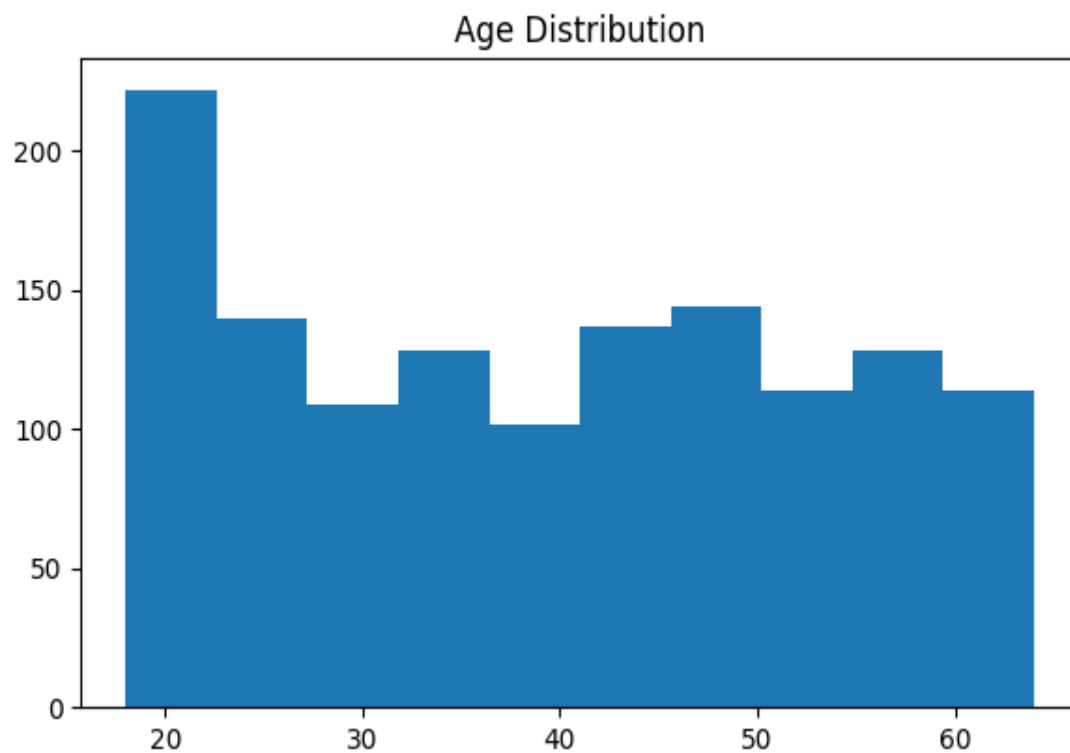
The dataset consists of customer demographic and medical information including age, gender, BMI, number of children, smoking status, region, and insurance charges.

3. Exploratory Data Analysis

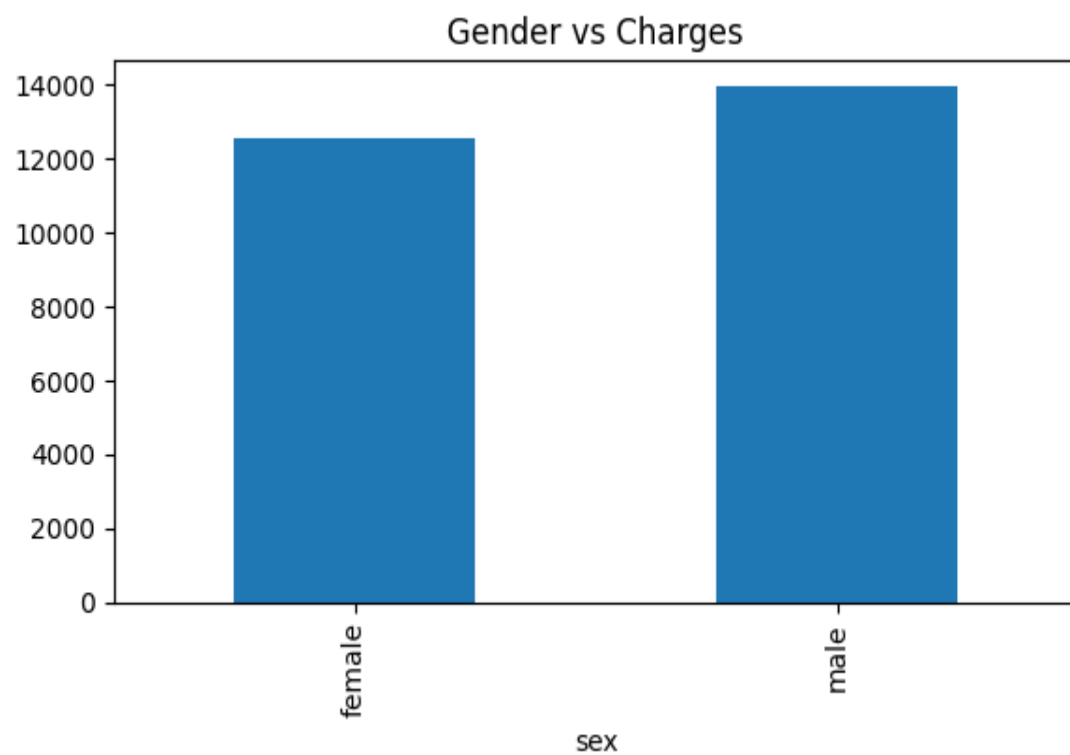
Insurance Charges Distribution



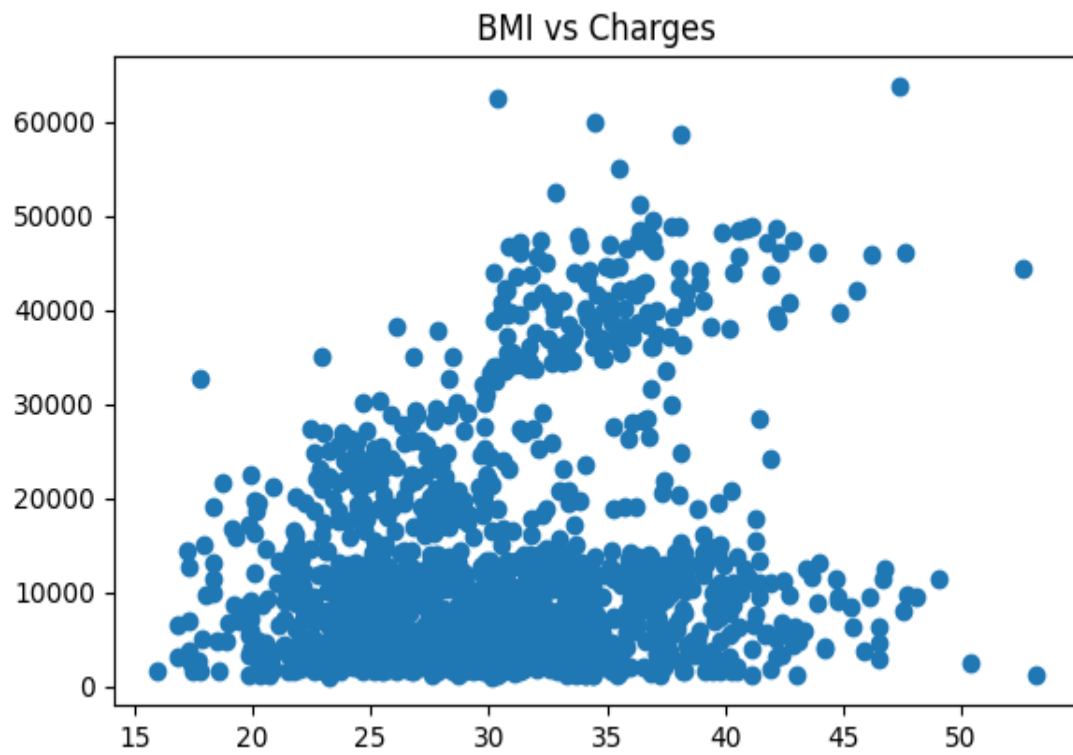
Age Distribution



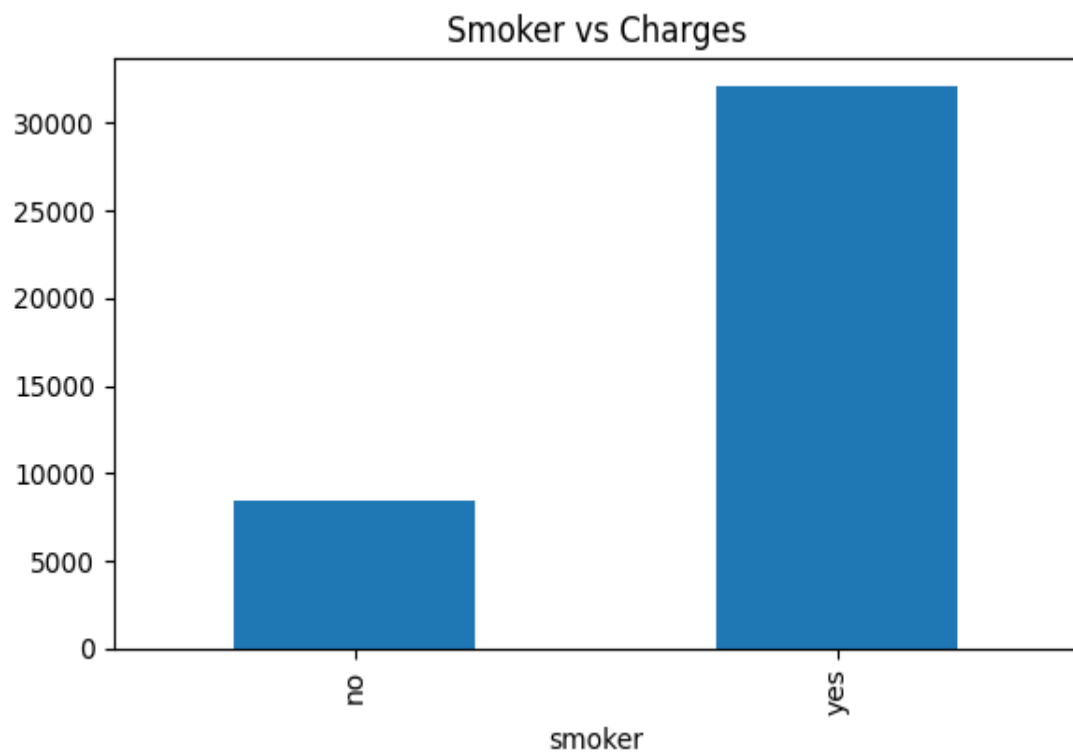
Gender vs Charges



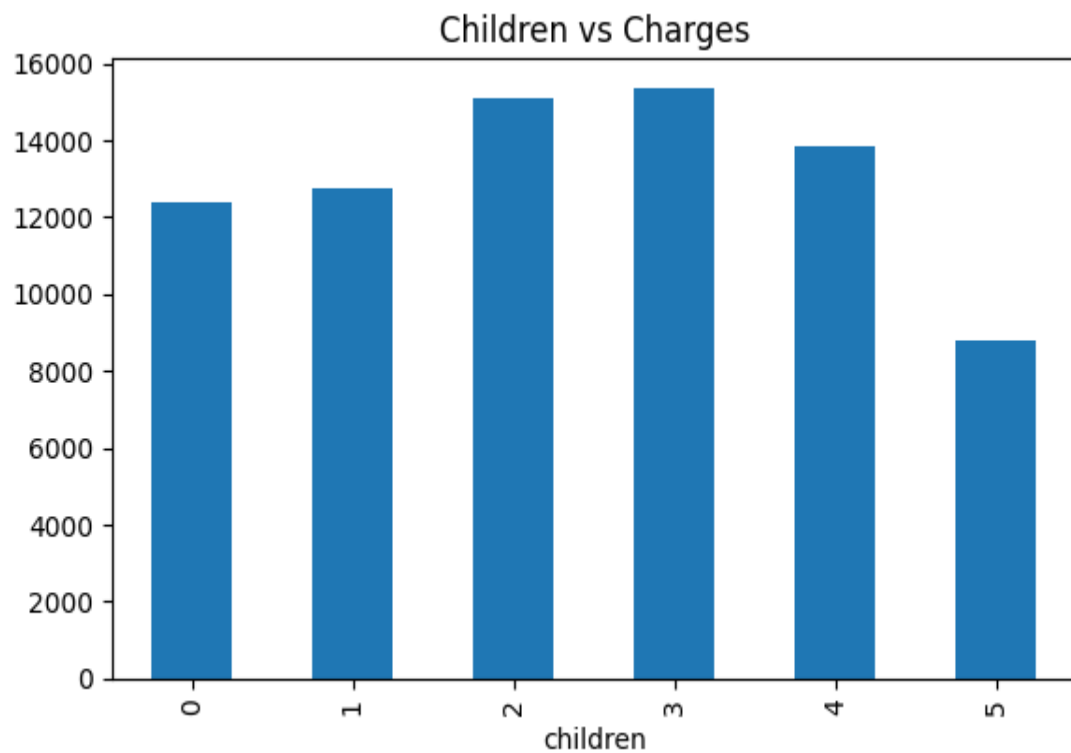
BMI vs Charges



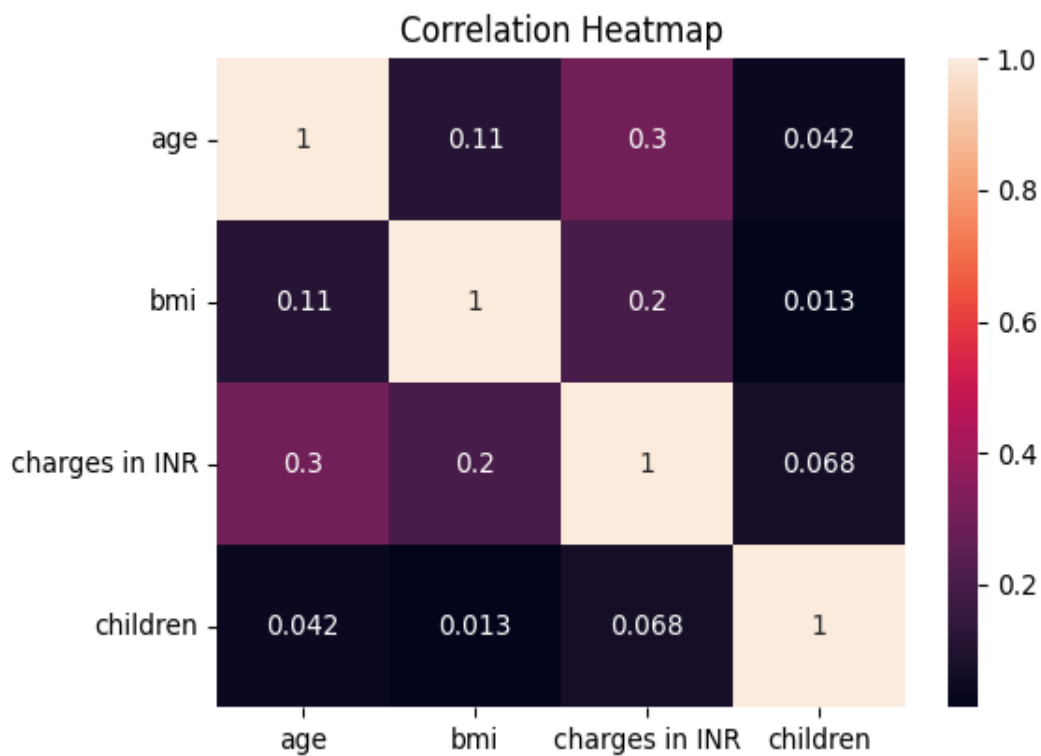
Smoker vs Charges



Children vs Charges



Correlation Heatmap



4. Key Insights

- Age, BMI, and smoking habits strongly affect insurance charges.
- Smokers incur significantly higher medical costs.
- Higher BMI leads to increased insurance expenses.
- Customers with more dependents have higher insurance claims.
- Gender and region have minimal impact on insurance pricing.

5. Conclusion

This analysis demonstrates how data-driven insights can help improve insurance pricing strategies. The findings enable MediBuddy to design better risk-based policies and improve decision-making.